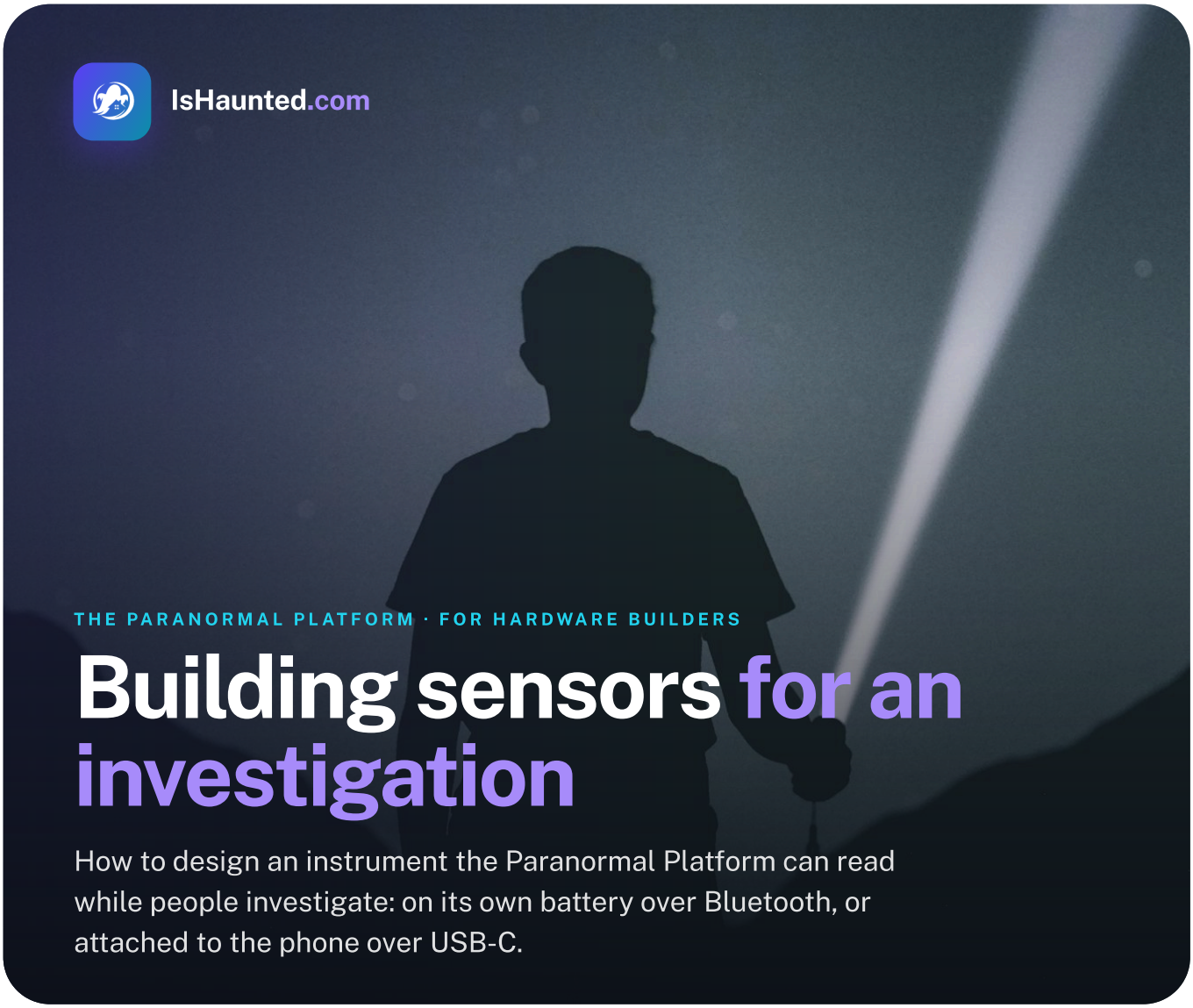


IsHaunted.com

THE PARANORMAL PLATFORM · FOR HARDWARE BUILDERS

Building sensors for an investigation

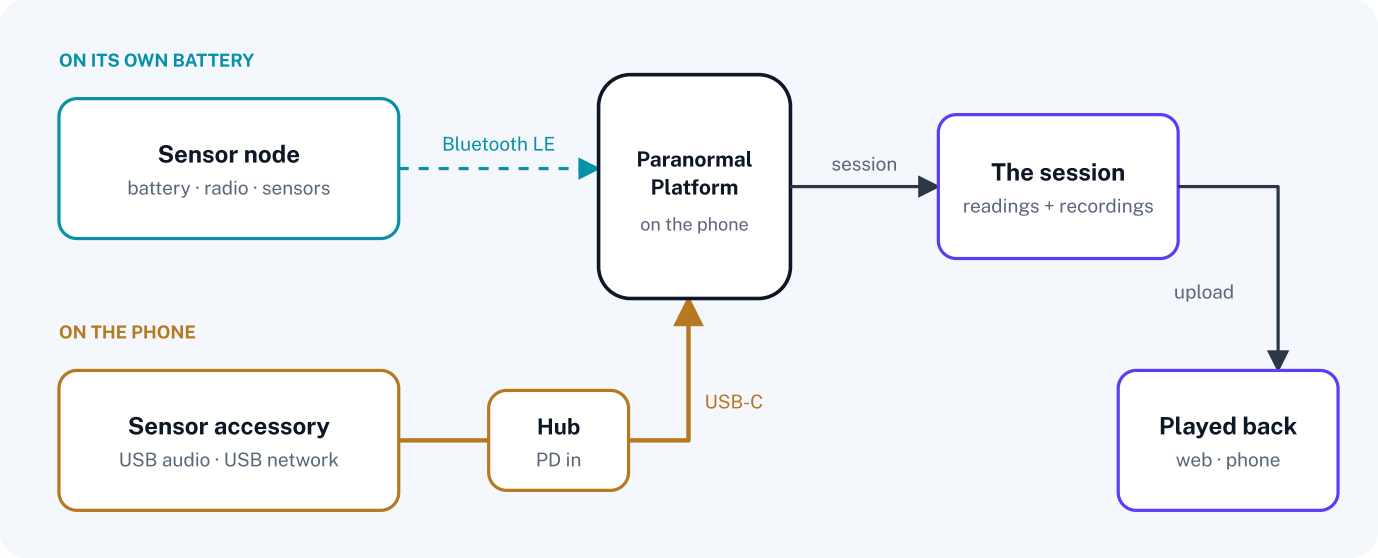
How to design an instrument the Paranormal Platform can read while people investigate: on its own battery over Bluetooth, or attached to the phone over USB-C.

WIRELESS	WIRED	DATA
Bluetooth LE	USB-C audio · network	Device Data Format v1

Who this is for: an electrical engineer designing new sensors. It covers what the phone can and cannot accept, a reference architecture for each route, the over-the-air contract, placement and noise, and how a reading ends up on the investigators' timeline. **The schematics are reference sketches** — starting points to design from, not validated circuits.

The short version

The Paranormal Platform turns a phone into an instrument panel: it reads the phone's own magnetometer, microphone, camera and location many times a second, keeps each reading against one clock, and plays a whole night back as one timeline. A sensor you build becomes one more channel on that timeline.



Two ways in. Either way, the phone places each reading on the session's clock, and everything downstream — the upload, the web player, the phone's own review — already knows how to carry a new channel.

Start with Bluetooth LE

A sensor on its own battery, talking Bluetooth LE. No Apple approval, works with every iPhone the app supports, and the sensor's own electronics stay away from the phone's compass. **Recommended for a first instrument.**

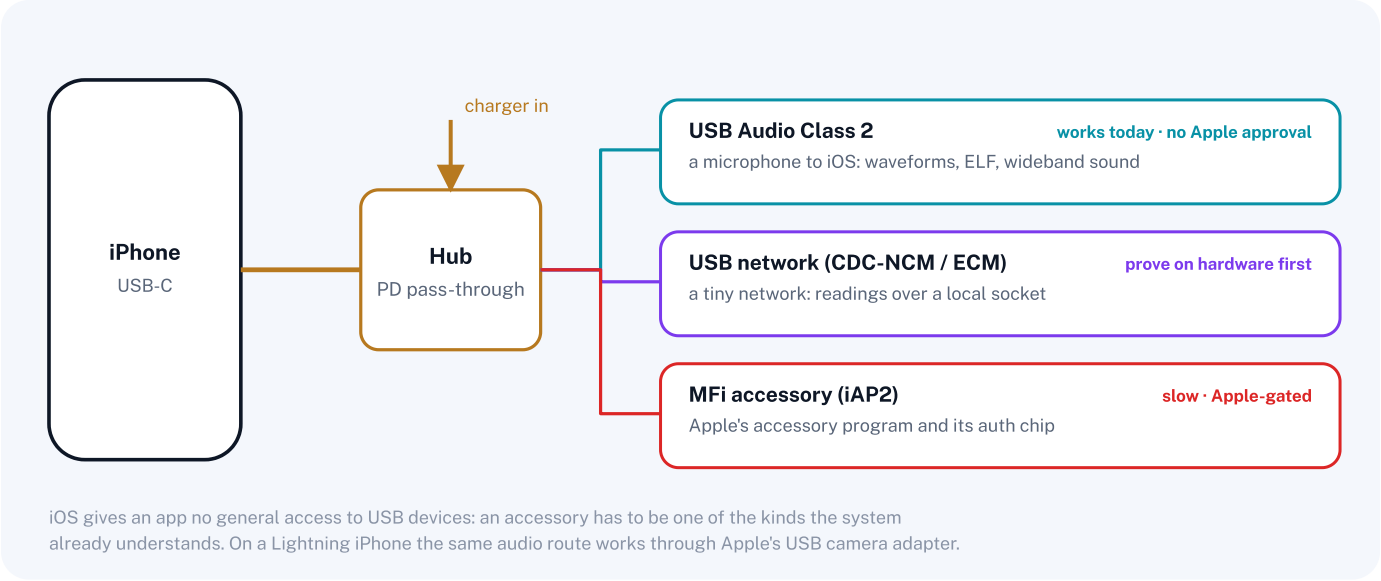
Wired, for waveforms

An accessory on the phone's USB-C port, through a hub that also charges the phone. Best when the signal is fast — an ELF waveform, wideband sound — and the accessory can present itself as a USB microphone.

What exists today, and what does not. The format your readings travel in, the upload, the timeline and both players exist and are built to take new channels. The phone's side of a new sensor — finding it, pairing, reading its channels, drawing its gauge — is built when the first instrument is ready to test against. This guide is the contract both sides build to.

What the phone will accept

An iPhone is not a USB host an app can program. iOS gives apps no general access to USB devices; an accessory has to be one of the kinds the system already understands. That narrows the wired choices to three, and makes Bluetooth LE the path of least resistance.



Wired routes. A hub with power-delivery pass-through keeps the phone charging through a long night while the accessory is connected.

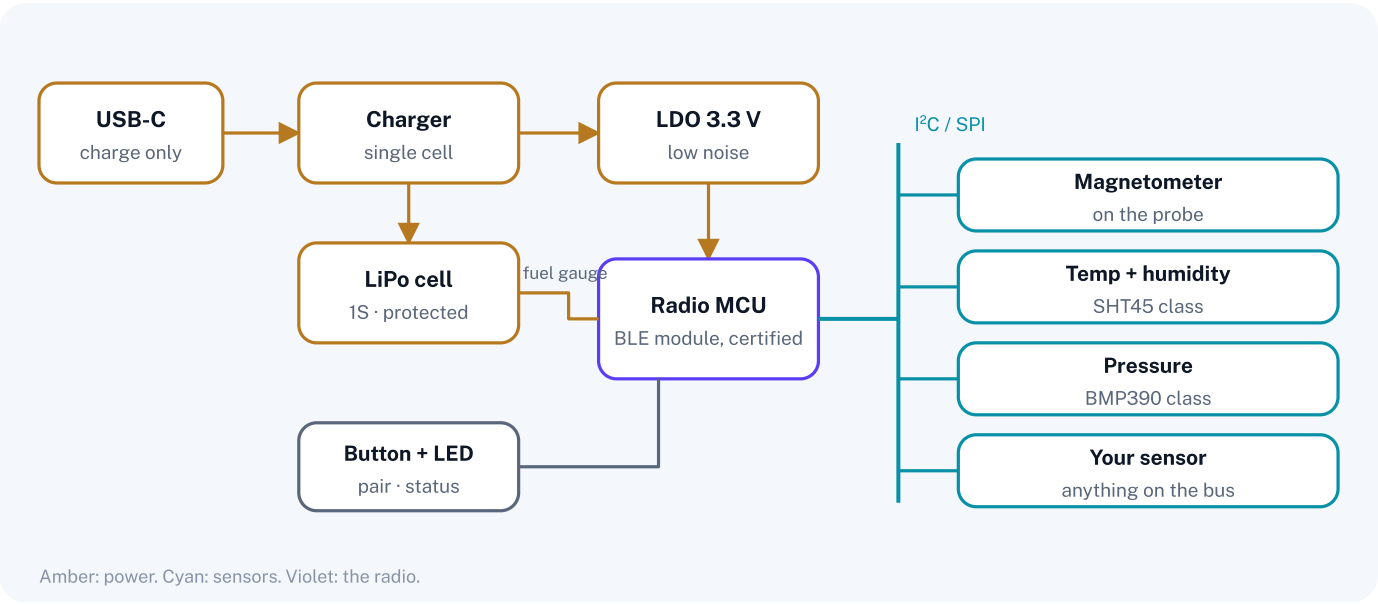
ROUTE	HOW THE APP READS IT	APPLE APPROVAL	GOOD FOR
Bluetooth LE	Core Bluetooth, as a GATT client	None	Slow-changing channels at up to tens of samples a second; anything on its own battery
USB Audio Class 2	The audio input, like an external microphone	None	Waveforms: ELF field, infrasound, ultrasonic, a better microphone
USB network (CDC-NCM/ECM)	A local socket to the accessory's address	None	Many channels at a high rate. **Prove on hardware first**: iPhones accept USB Ethernet adapters, but test a custom device early
MFi accessory	External Accessory framework (iAP2)	Apple's MFi program + authentication chip	A volume product, later. Not for a first build

Phones with Lightning (iPhone 14 and earlier) reach the same USB audio route through Apple's Lightning to USB 3 Camera Adapter, which also passes charging power. Bluetooth LE works on all of them.

Power from the phone: an iPhone's USB-C port can power a small accessory (Apple quotes 4.5 W for charging other devices). Budget well under that, and expect the phone to cut an accessory that draws too much at connection.

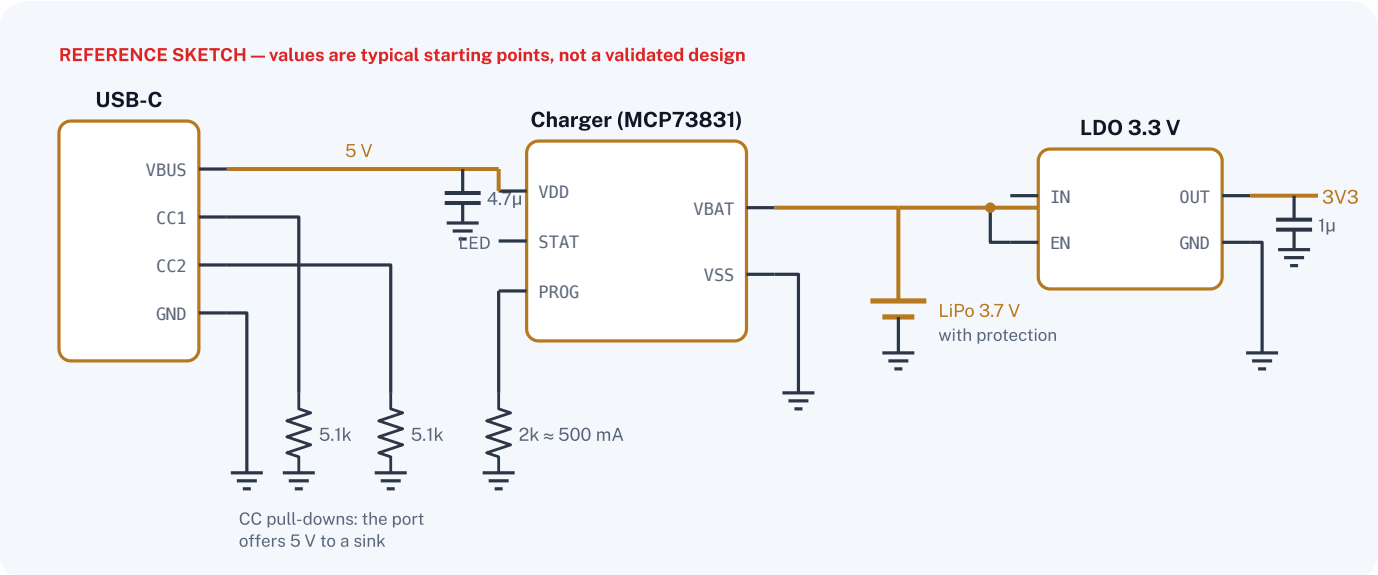
A Bluetooth sensor node

A single-cell lithium battery, a certified radio module, a quiet regulator and the sensors on one bus. Use a **pre-certified BLE module** (nRF52840 or ESP32-S3 class) rather than a bare radio chip: the module's certification covers the transmitter, which is the expensive part of getting a product to market.



Reference architecture. A fuel gauge lets the node report its own battery, which the format records because a low battery is a known cause of false readings on many meters.

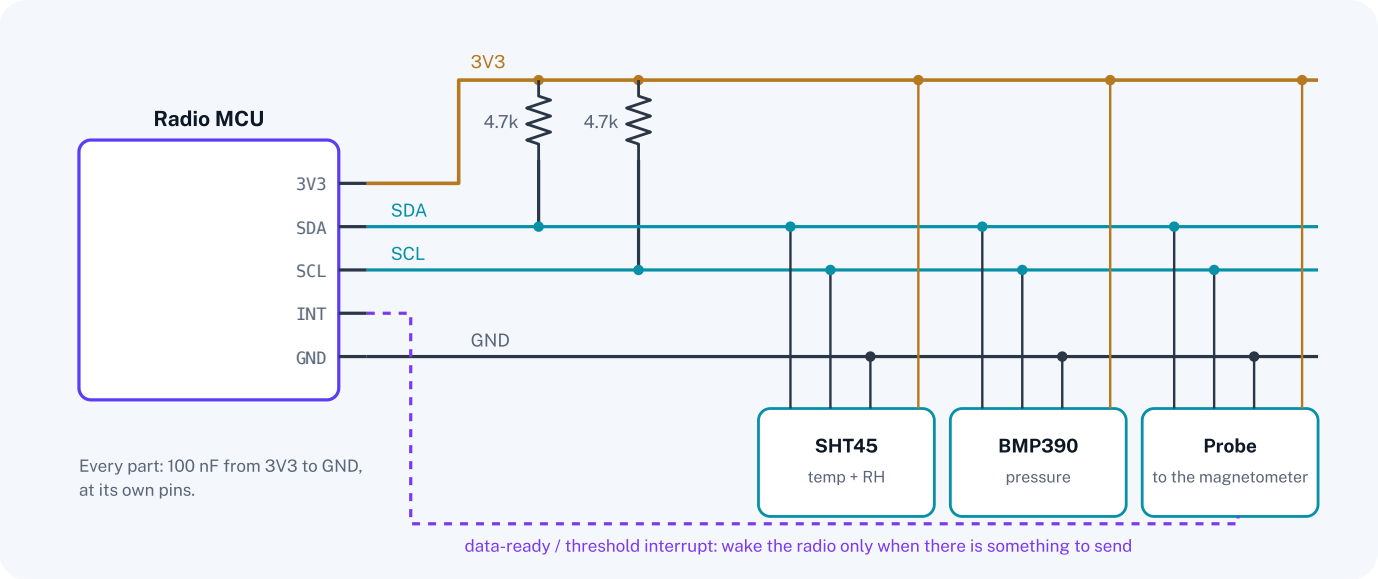
Power



Charging from USB-C as a sink (5.1 kΩ on each CC pin), a single-cell linear charger, and a low-noise LDO. A linear regulator wastes a little battery but adds no switching noise next to the sensors.

- **Battery life target: five hours or more.** The phone's own recording is designed to last an investigation of that length, and a sensor that dies first leaves a gap that reads like silence.
- **Prefer an LDO near magnetic sensors.** If a switching regulator is unavoidable, put it at the far end of the board, keep its inductor's loop small, and measure the node's noise floor with the regulator running.
- **Report the battery honestly** over the standard Battery service, and the time since power-on: several instruments drift until warm, and the format records both for that reason.

The sensor bus



One I²C bus with 4.7 kΩ pull-ups and 100 nF on every part. A data-ready or threshold interrupt lets the radio sleep until there is something worth sending.

Over the air: the contract

Use the standard Bluetooth services wherever one fits — any BLE tool can read them, which makes bring-up and debugging much easier — and one IsHaunted service for everything else.

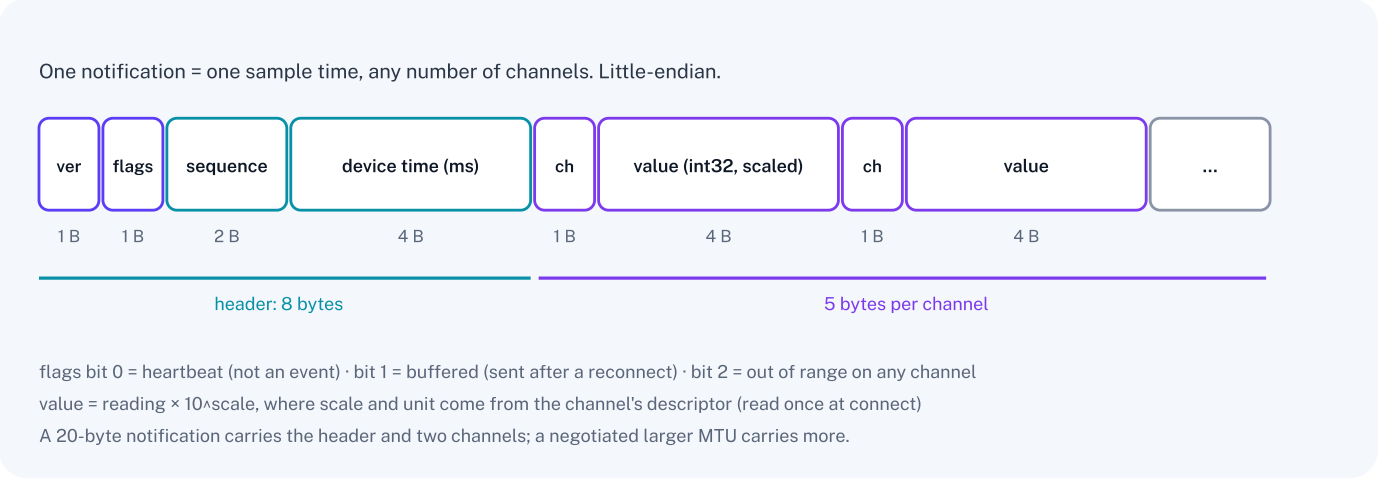
SERVICE	UUID	CHARACTERISTICS	ACCESS	BECOMES
Device Information	`0x180A`	Manufacturer 0x2A29 · Model 0x2A24 · Serial 0x2A25 · Firmware 0x2A26	read	The session's `device` block, word for word.
Battery	`0x180F`	Battery Level 0x2A19 (uint8, %)	read · notify	`battery_percent_at_start`, and the `battery` channel.
Environmental Sensing	`0x181A`	Temperature 0x2A6E · Humidity 0x2A6F · Pressure 0x2A6D · Magnetic Flux Density 3D 0x2AA1	read · notify	Use the standard ones where they fit: any BLE tool can read them.
IsHaunted Sensor	`8bda9059-8022-458e-87f1-1ffc5f0e8208`	the characteristics in the next table	—	Everything the standard services cannot say.

The IsHaunted Sensor service

CHARACTERISTIC	UUID	ACCESS	CARRIES
Readings	`8bda0001-8022-458e-87f1-1ffc5f0e8208`	notify	The packet above, one per sample time.
Channels	`8bda0002-8022-458e-87f1-1ffc5f0e8208`	read	JSON: each channel's id, name, unit, scale, accuracy and range.
Control	`8bda0003-8022-458e-87f1-1ffc5f0e8208`	write	Start, stop, rate, set base level, report-at threshold.
Clock	`8bda0004-8022-458e-87f1-1ffc5f0e8208`	read	The device's millisecond counter now (uint32), for the first offset.
Status	`8bda0005-8022-458e-87f1-1ffc5f0e8208`	read · notify	Warm-up done, calibration date, faults, buffered count.

Service `8bda9059-8022-458e-87f1-1ffc5f0e8208` is assigned to this project. Advertise it, so the phone can list only IsHaunted-compatible sensors when someone pairs one.

A reading on the wire



The Readings characteristic. A sequence number lets the phone tell a missed sample from a quiet one; the device time places each sample exactly, however late the radio delivered it.

The Channels descriptor

Read once at connection. It is what turns channel 3 and an integer into "1.24 µT, ±0.05":

```
{
  "format": 1,
  "channels": [
    { "id": 1, "name": "emf_x", "unit": "uT", "scale": -2, "accuracy": 0.05, "range": [-800, 800] },
    { "id": 2, "name": "emf_y", "unit": "uT", "scale": -2, "accuracy": 0.05, "range": [-800, 800] },
    { "id": 3, "name": "emf_z", "unit": "uT", "scale": -2, "accuracy": 0.05, "range": [-800, 800] },
    { "id": 4, "name": "temperature", "unit": "degC", "scale": -2, "accuracy": 0.2 },
    { "id": 5, "name": "elf_field", "unit": "uT", "scale": -3 },
    { "id": 6, "name": "elf_frequency", "unit": "Hz", "scale": 0 }
  ],
  "trigger": { "mode": "hybrid", "interval_seconds": 2, "event_description": "field exceeds 2.0 uT
above base" }
}
```

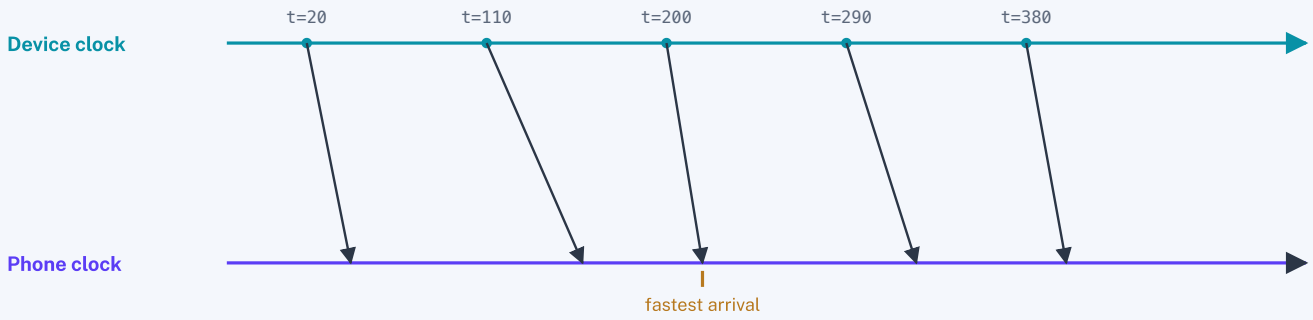
Names follow the format's conventions — lowercase, `snake_case`, ASCII units (`uT`, not µT). A channel the platform has never seen is still recorded and kept; it is drawn once someone adds a gauge for it, so a new instrument is never blocked waiting on the app.

Connection settings Apple asks for

- **Connection interval:** at least 15 ms. Apple's accessory guidelines set the exact rules (minimums, multiples, latency and supervision timeout) — design to the current edition of *Accessory Design Guidelines for Apple Devices*, Bluetooth chapter.
- **Advertising:** fast (about 20 ms) for the first 30 seconds after power-on or disconnect, then one of the longer intervals Apple lists, to save battery.
- **Packet size:** design so the header and two channels fit 20 bytes; if the phone negotiates a larger MTU, send more channels per notification.
- **Pairing:** "Just Works" is fine for a measuring instrument; nothing secret crosses this link.

Time is the whole point

An investigation is read by lining things up: the field jumped at the moment the recording caught a knock. That only works if every reading carries the right time. The node keeps its own millisecond counter; the phone works out how that counter maps to its clock.




```
offset = min over a window of (phone arrival − device time) — the least-delayed sample is the honest one
drift = the slope of that offset over the session — a 20 ppm crystal drifts 72 ms an hour
at = device time + offset(at that moment) — each reading keeps its own time, even when buffered
```

Bluetooth delivers notifications late by a varying amount. The least-delayed one in a window gives the true offset; tracking it over the session corrects the node's crystal drift.

- **Stamp at the moment of measurement**, not when the packet is sent.
- **Never reset the counter during a session.** A wrap of a 32-bit millisecond counter takes 49 days; a reset mid-session is what breaks alignment.
- **Buffer through a dropped connection** and send the backlog afterwards with its original times, flagged as buffered. A reading that arrives late but correctly timed is worth exactly as much as one that arrived on time.
- **Say how precise you are.** The format records a precision for each reading, so a reviewer never reads false accuracy into a timestamp. Aim for better than 50 ms against the phone's clock.

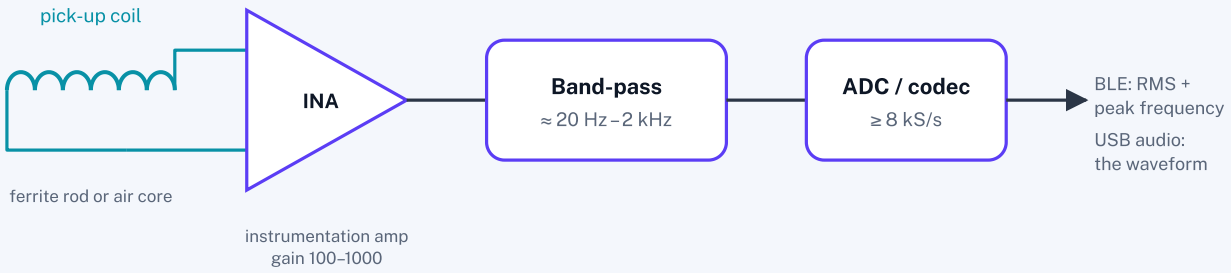
Sensors worth building

What investigators reach for, the parts that do it well, and the channel name each should report as. Part numbers are starting points from widely used families — check current datasheets for noise and range before committing.

MEASURES	CHANNEL	UNIT	CANDIDATE PARTS	BUS	NOTES
Magnetic field, 3-axis	emf, emf_x/y/z	uT	LIS3MDL, MMC5983MA, RM3100	I²C/SPI	The core instrument. RM3100-class parts are far quieter than a phone's compass. Calibrate hard and soft iron per unit.
ELF (AC) magnetic	elf_field + elf_frequency	uT, Hz	coil + INA333/INA128 + ADC or codec	analog	What hand-held EMF meters actually show. Report level and the strongest frequency.

MEASURES	CHANNEL	UNIT	CANDIDATE PARTS	BUS	NOTES
Electric field (static / ELF)	e_field	relative	antenna + electrometer op-amp (LMP7721)	analog	Very high impedance: guard the input trace. Report change from base, not an absolute.
Temperature + humidity	temperature, humidity	degC, pct	SHT45, BME280	I²C	Ventilate the sensor away from the battery and regulator, which warm it.
Air pressure	pressure	hPa	BMP390, LPS22	I²C	Doors, drafts and floors: small, fast changes are the interesting part.
Surface temperature (non-contact)	ir_temperature	degC	MLX90614, MLX90632	I²C	Cold spots on walls and floors; note the field of view.
Thermal array	thermal_min / thermal_max	degC	AMG8833 (8×8), MLX90640 (32×24)	I²C	Send min/max/mean as readings; a frame is a picture, not a reading (attach it as a file).
Light	illuminance	lx	VEML7700, TSL2591	I²C	Flashlights, flickers, a door opening onto a lit hall.
Vibration / footfalls	vibration	mm/s or g	geophone SM-24 + ADS1220, or ADXL355	SPI/analog	A geophone hears footsteps in another room; mount it on the floor, not the table.
Radiation	radiation	uSv/h or cpm	PIN-diode module (BG51), or a GM tube	pulse	A GM tube needs ~400 V: see the safety note. The PIN-diode route avoids high voltage.
Presence (radar)	presence	bool + m	60 GHz mmWave modules	UART	Sees movement through thin walls. It is a transmitter: keep it off during EMF work.
Passive infrared	motion_detected	bool	Panasonic EKMC series	GPIO	Cheap, reliable, and exactly what the format's suggested name expects.

The ELF front end



Mains (50/60 Hz) and its harmonics dominate indoors. Report the level AND the strongest frequency, so a reviewer can tell house wiring from anything else.

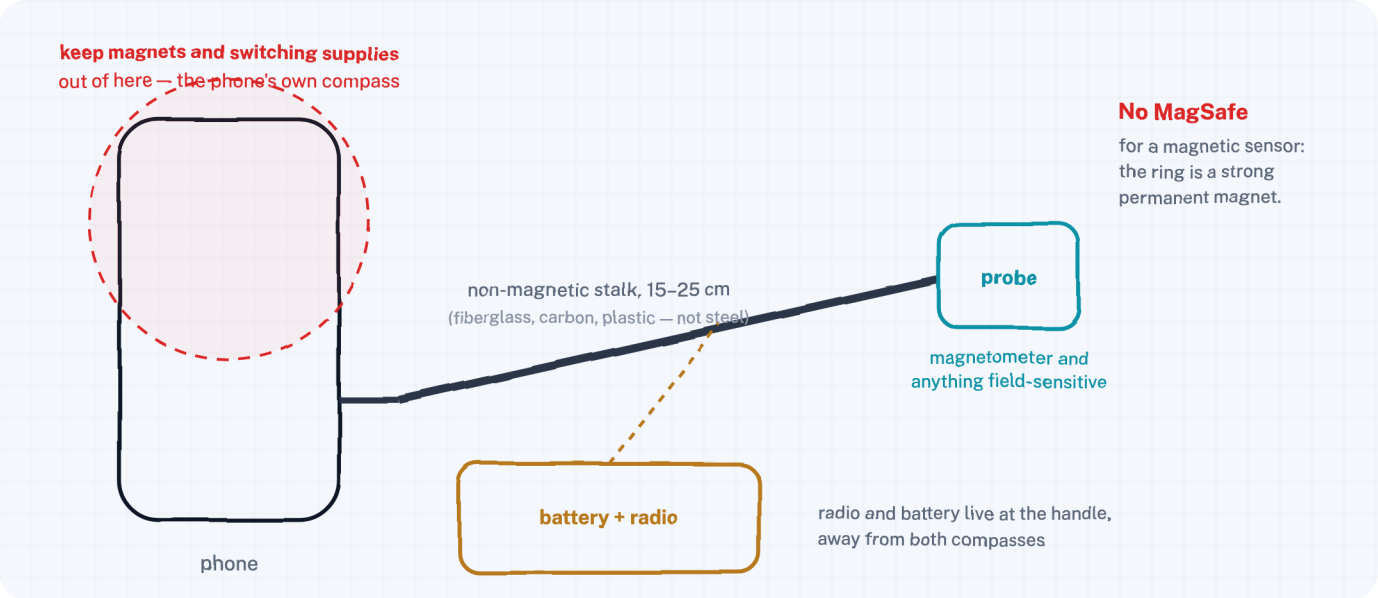
A pick-up coil, an instrumentation amplifier and a band-pass. Over Bluetooth, send the RMS level and the strongest frequency a few times a second; over USB audio, send the waveform itself.

High voltage. Geiger-Müller tubes run at around 400 V. Build that supply with creepage distances, a bleeder, a current-limited converter and an enclosure that cannot be opened while charged — or use a PIN-diode radiation module and avoid high voltage entirely.

Transmitters. A radar presence sensor, and the node's own radio, are sources of the very fields an EMF channel measures. Let the Control characteristic turn the radar off, and measure the node's own signature so it can be told apart.

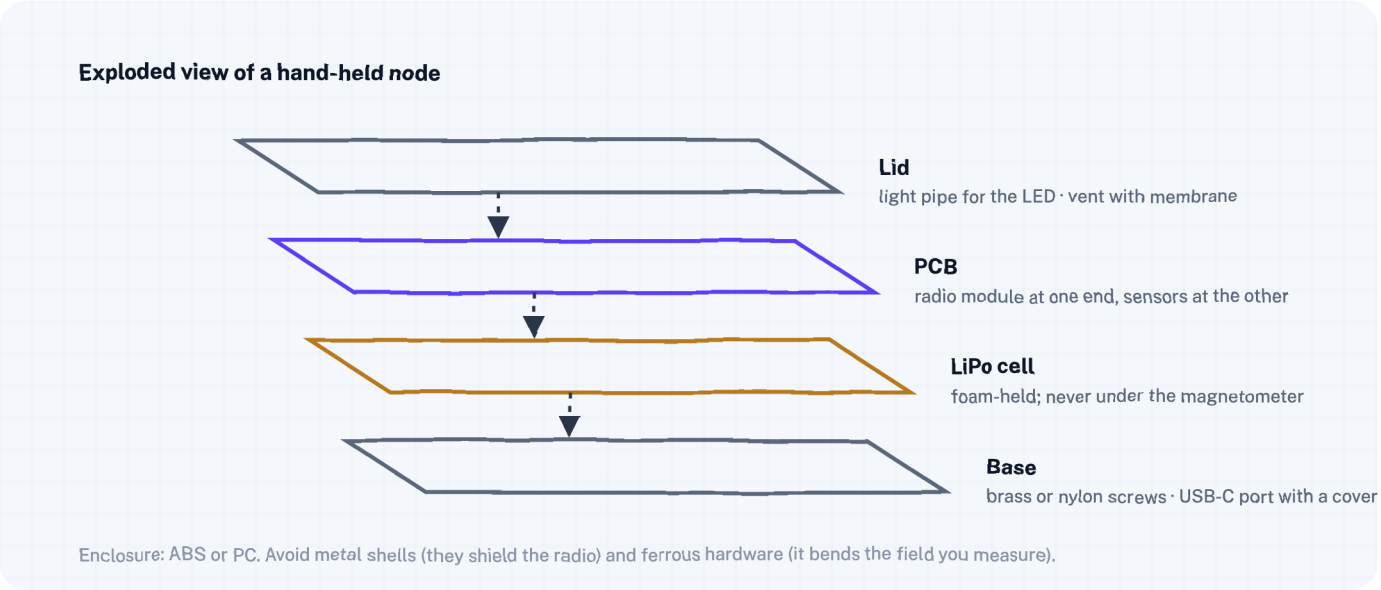
Placement and noise

A magnetic instrument mostly measures whatever is closest to it. The phone, the node's battery, its regulator and its radio are all closer than the room.



Sketch: keep the probe on a non-magnetic stalk, the battery and radio at the handle, and nothing magnetic near the phone.

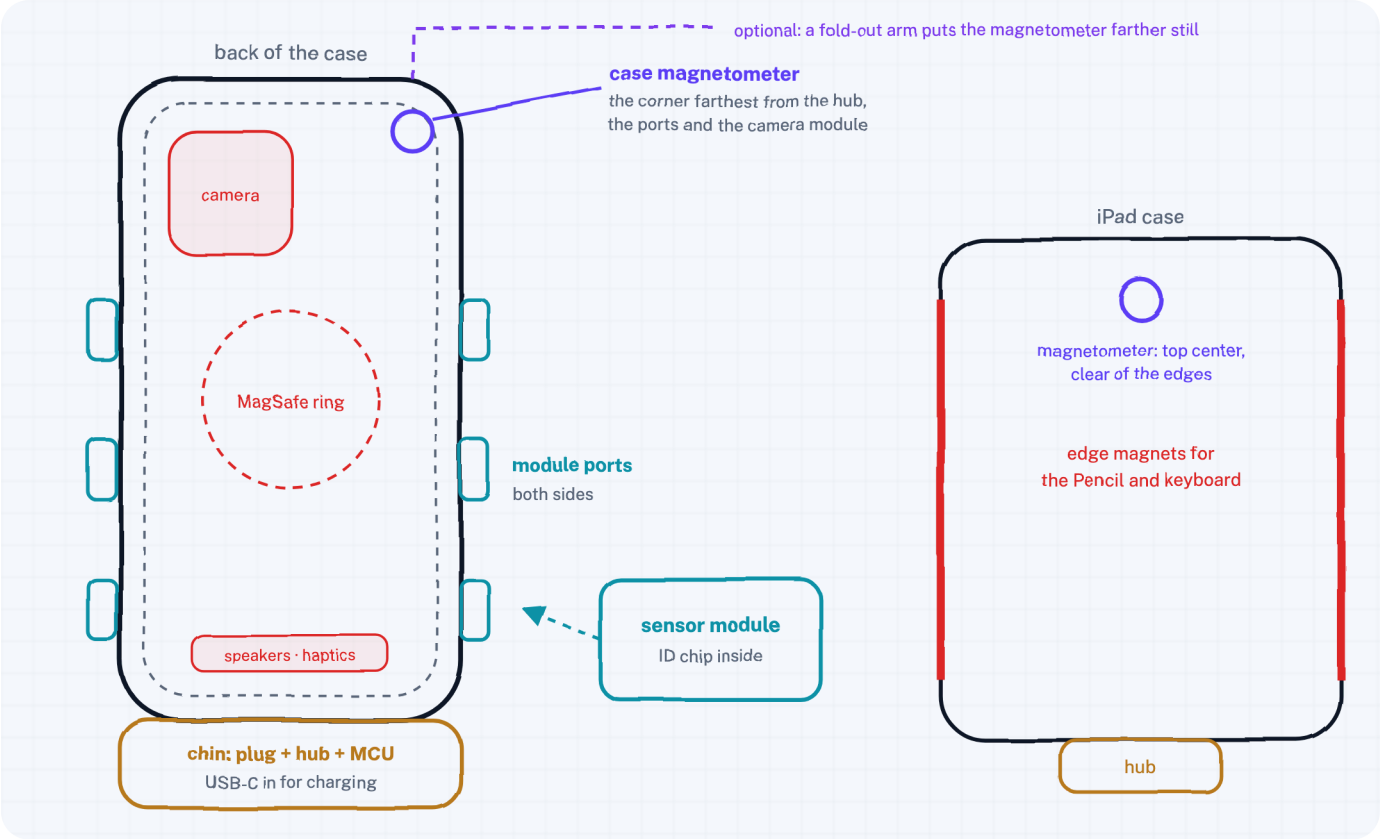
- **Measure the node's own floor** in a quiet place, radio transmitting, regulator running: that number belongs in the Status characteristic and in the instrument's documentation.
- **Calibrate magnetometers per unit** — hard-iron offsets and soft-iron scale — and store the result on the node with its date. The phone records it with every session.
- **Baseline is the investigators' job**, and the Control characteristic is how they do it: "set base level" on the phone asks every connected sensor to take its own.
- **Mark a pegged sensor as out of range** rather than sending its maximum. The format has a field for it, because a clipped maximum read as a measurement is a false finding.



Sketch: a hand-held node in four layers. Nylon or brass hardware; the battery held away from the magnetometer end.

A sensor case for iPhone and iPad

One idea under consideration: a case that plugs into the phone's USB-C port like a hub, with ports along its sides for sensor modules, so an investigator clips on the instruments the night needs. The same idea works for an iPad, with more room and different hazards.



Sketch: the hub and its processor live in the chin, modules plug into the sides, and the case's own magnetometer sits in the corner farthest from all of them — and from the phone's own magnets.

How the case reaches the phone

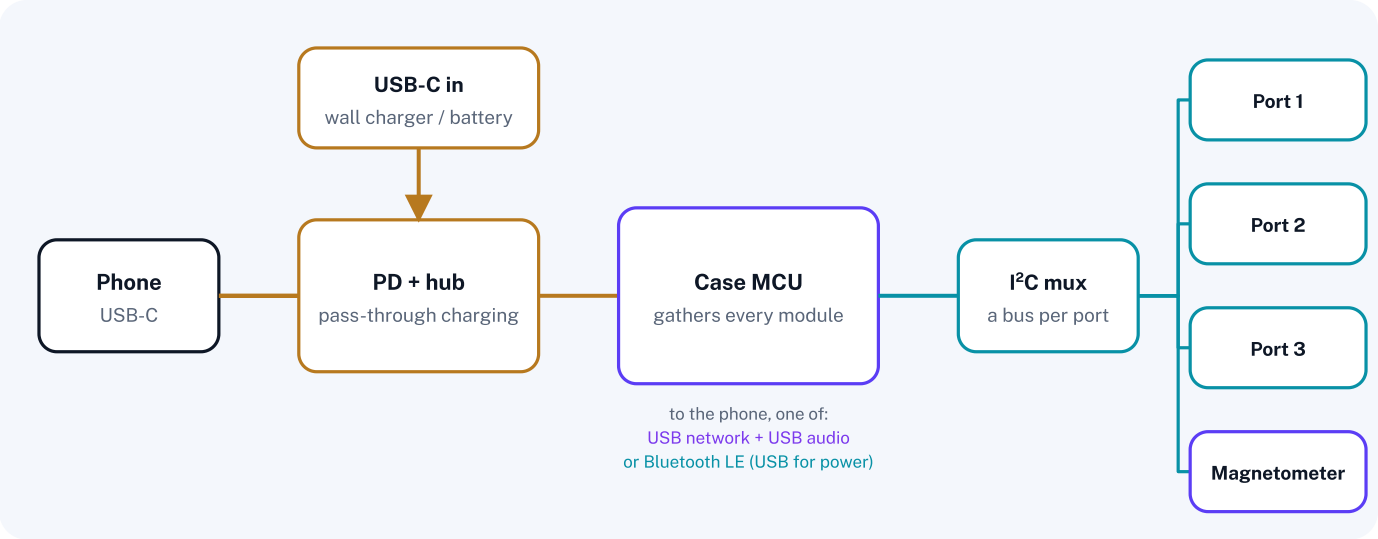
The modules' bus cannot be handed to iOS as it is; a small processor in the case reads every module and speaks to the phone as one device, by one of the routes under *What the phone will accept*:

All wired

The case is a USB composite device: a USB network interface for the readings, and optionally a USB audio interface for a waveform channel. One cable, no radio — but the network route must be proven on an iPhone early.

Wired power, Bluetooth data

USB only powers the case and passes charging through; readings go over Bluetooth LE exactly as from a battery node. It depends on nothing iOS might refuse, and the same firmware serves the case and a stand-alone node.



Case electronics. An I²C multiplexer gives each port its own bus, so two modules built around the same part — and the same address — can be fitted at once.

Modules that describe themselves

Each module carries a small EEPROM holding its Channels entry (name, unit, scale, accuracy), its serial and its calibration. The case reads it when a module is plugged in and adds those channels to what it reports, so a new module needs no new case firmware. A suggested port:

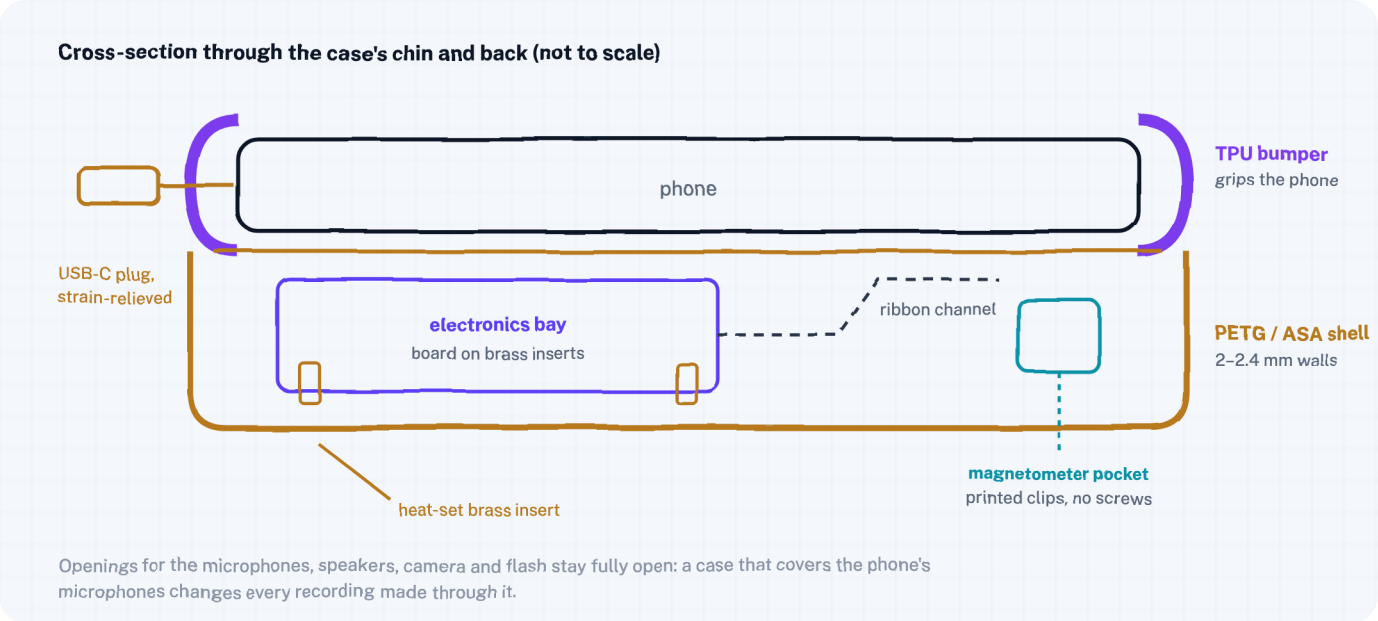
PIN	SIGNAL	NOTES
1	3V3	From the case, current-limited per port (a load switch with a fault flag)
2	GND	Make first, break last: the longest pin
3	SDA	Through the multiplexer; the module has no pull-ups of its own
4	SCL	As SDA
5	INT	Open-drain, shared: data ready or threshold crossed
6	DETECT	Pulled low by the module, so the case sees it arrive and leave

The magnetometer in the case

- **The phone is a magnet.** Speakers, the vibration motor, the camera's stabilizer and the MagSafe ring all contain permanent magnets. Their steady field is removed by calibrating the case magnetometer **with the phone fitted** — and recalibrating if the phone changes.
- **What calibration cannot remove** is what changes: the speaker playing, the motor buzzing, the phone charging. Record those moments (the phone knows when it is playing sound) so a reviewer can set them aside.
- **An iPad's edges are lined with magnets** for the Pencil and the keyboard. Put the magnetometer top center, clear of the edges, and do not design an iPad case that holds a Pencil magnetically.
- **Keep the hub's switching parts in the chin,** the modules' current away from the magnetometer corner, and measure the case's floor with every port loaded. If that floor is too high, a fold-out arm puts the magnetometer farther away still.

If the case is 3D printed

Printing is the right way to iterate on a case: a new revision in a day, and every phone model is a new file rather than a new tool. What changes for an investigation case:



Sketch: a flexible TPU bumper that grips the phone, a rigid shell that carries the electronics, brass inserts for the board, and the magnetometer in a pocket of its own at the far end.

MATERIAL	USE IT FOR	WATCH OUT FOR
PETG	The rigid shell and module housings	Strong and slightly flexible; prints easily. Softens around 75 °C
ASA	The shell, for a case that lives in a car	UV- and heat-resistant (around 95 °C); needs an enclosed printer
TPU (95A)	The bumper that grips the phone	Flexible; slow to print. Print the bumper separately, or as a second material
Nylon (SLS / MJF, PA12)	A production run from a print service	Tough, even, no layer lines; the step after FDM prototypes
PLA	Fit checks only	Too brittle for clips, and softens around 55 °C — a dashboard in summer
Iron- or steel-filled	Nothing	Never: it is magnetic, and turns the case into part of every reading

- **Carbon-fiber-filled filaments are not magnetic**, so they are safe near the magnetometer, but they are slightly conductive and dull a Bluetooth antenna placed against them. Keep the radio's antenna end in plain plastic.
- **Fit:** work from Apple's published dimensional drawings for each phone and iPad (in the Accessory Design Guidelines), not from measuring one phone. Allow about 0.2–0.3 mm clearance per side for the rigid shell and let the TPU bumper do the gripping; print a fit test of just the corners before a whole case.
- **Walls and clips:** 2–2.4 mm walls for the shell. Print so layer lines run *along* a snap-fit clip, not across it, or it snaps at the first fitting.

- **Fasteners:** heat-set **brass** inserts and brass or nylon screws. A steel screw beside the magnetometer is a hard-iron offset that changes when it is tightened.
- **Port contacts:** spring-loaded pogo pins usually have steel springs. Use ones with non-magnetic (beryllium copper) springs on any port near the magnetometer, and put a TVS diode on every exposed contact: a plastic case gives static discharge nowhere else to go.
- **No magnets** to hold modules on. A mechanical latch or a printed snap does the same job without adding a field.
- **Openings:** the phone's microphones, speakers, camera, flash and buttons stay fully open. A case over a microphone muffles every recording made through it — exactly the evidence the platform exists to keep.
- **Heat:** a phone recording video for hours runs warm, and plastic insulates it. Leave the back over the phone's hottest area open or vented, and keep the battery bay away from it.

Power

Pass-through charging is what makes the case usable for a whole night: a USB-C input on the chin feeds a power-delivery controller that charges the phone and powers the modules. Without a charger, the modules run from the phone's port, within its small power budget — or from a cell in the case, which also keeps the phone's battery for the recording itself.

From a sample to the player



What the builder owns: the first two boxes, and the descriptor that names each channel.
What the platform owns: everything after the phone receives a packet.

The builder's part ends when the phone has the packet. From there a reading is stored in the session, uploaded with it, and drawn on the same timeline as the recordings and the map.

Inside the session, a reading from your node looks like this — one more key under `measurements`, beside what the phone measured itself:

```
{
  "at": "2026-10-31T23:14:07.240Z",
  "precision": "millisecond",
  "sequence": 5812,
  "triggered_by": "event",
  "measurements": {
    "emf":           { "value": 0.62, "unit": "uT", "baseline": 0.31 },
    "probe_emf":     { "value": 1.24, "unit": "uT", "accuracy": 0.05, "baseline": 0.40 },
    "elf_field":     { "value": 0.180, "unit": "uT" },
    "elf_frequency": { "value": 60, "unit": "Hz" },
    "temperature":   { "value": 17.42, "unit": "degC", "accuracy": 0.2 }
  }
}
```

The rules that make this safe to extend: every number carries a unit; everything but the time is optional; a reader ignores channels it does not know rather than rejecting the file. A new instrument is an additive change — the format's minor version moves, nothing breaks.

Before a prototype comes to a test night

- □ Advertises the IsHaunted service and answers Device Information with real names and a firmware version.
- □ Channels descriptor reads back valid JSON with a unit on every numeric channel.
- □ Sequence numbers never repeat and never skip, except across a genuine loss.
- □ Survives walking out of range and back: buffered readings arrive with their original times.
- □ Runs five hours from a full charge with the radio connected.
- □ Noise floor measured with the radio and regulator running, and written down.

- ☐ Warm-up drift measured over the first 30 minutes after power-on.
- ☐ Reports out-of-range rather than a clipped maximum.
- ☐ Nothing magnetic in the enclosure, and no MagSafe mount on a magnetic sensor.
- ☐ For high voltage: enclosure closed, supply current-limited, bleeder fitted.

What to send us with a prototype

The Channels descriptor, the GATT table as built, a capture of a few minutes of notifications (any BLE sniffer or phone tool), the measured noise floor and battery life, and the calibration procedure. With those, the phone's side can be built and tested against your instrument rather than an idea of it.